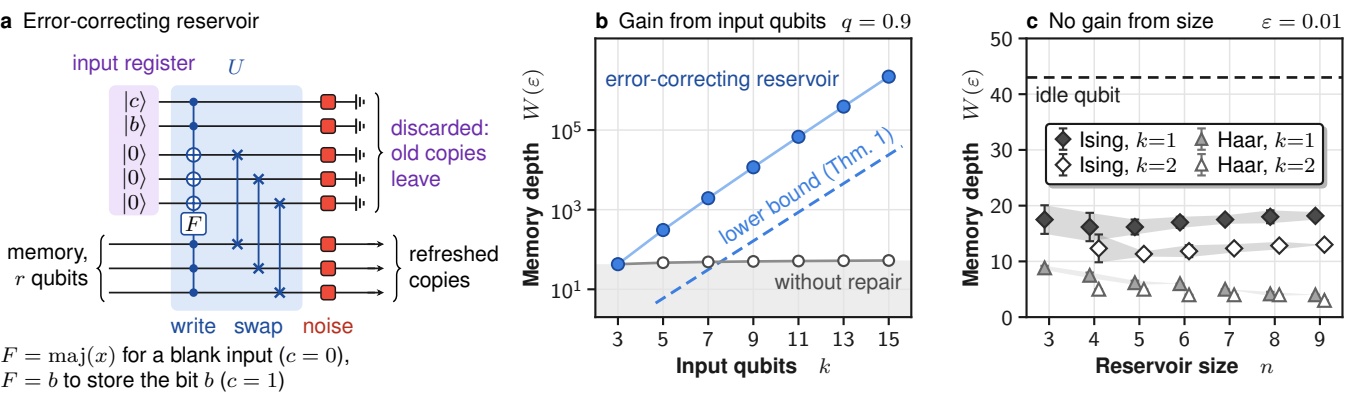

a Error-correcting reservoir
input register U
$|c\rangle$
$|b\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
F
discarded: old copies leave
memory, r qubits
refreshed copies
write
swap
noise
$F = \mathrm{maj}(x)$ for a blank input ($c = 0$),
$F = b$ to store the bit b ($c = 1$)
b Gain from input qubits $q = 0.9$
error-correcting reservoir
lower bound (Thm. 1)
without repair
Memory depth $W(\varepsilon)$
10^5
10^3
10^1
3 5 7 9 11 13 15
Input qubits k
c No gain from size $\varepsilon = 0.01$
idle qubit
Ising, k=1
Ising, k=2
Haar, k=1
Haar, k=2
Memory depth $W(\varepsilon)$
50 40 30 20 10 0
3 4 5 6 7 8 9
Reservoir size n